Coupling discrete Turing reaction–diffusion to a 3D vertex model:
what transfers cleanly, and what needs force balance

Notes from a `plexus2` prototype of Okuda et al. (2018)

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Abstract

We report the lessons from building a `plexus2` prototype that couples a *discrete* Turing reaction–diffusion (RD) system to a 3D vertex / self-propelled-Voronoi model on a growing epithelial vesicle, following Okuda *et al.*, *Sci. Rep.* **8**:2386 (2018). Four findings stand out. **(1)** When growth is morphogen-*independent*, the RD is a passive *colouring* of the cells: it never touches the mechanics. Apparent "noise" in Gray–Scott patterns was therefore not a chemistry problem but a *timestep* artefact — the coarse step Gray–Scott needs to develop ($\Delta t = 0.35$) destabilises the vertex mechanics, which had to be detuned to stay stable. Running the mechanics at a fine, stable step ($\Delta t = 0.02$) while *time-rescaling* the RD by a single rate multiplier recovers a clean epithelium carrying the coral / labyrinth / holes patterns (cell-shape roughness 0.04 vs. 0.24). This is exactly the chemistry-vs-mechanics timescale separation ($\Delta t_m \ll \Delta t_v$) of the original model. **(2)** A latent operator bug — a missing `SphericalVoronoi` import silently swallowed by a broad `except` — had made the tangential (Lloyd) relaxation a *total no-op* for the operator's entire life; the cell uniformity seen in earlier renders came *entirely* from a render-side cosmetic, which masked the dead physics. **(3)** Reproducing morphogen-driven *tubulation* (their Fig. 5) is qualitatively harder: an overdamped, single-step vertex relaxation *without* force-balance iteration or topological reconnection converts localised growth into *global* chaotic buckling, uncorrelated with the morphogen ($\mathrm{corr}(a, r) \approx 0$), rather than local tubes. **(4)** A working alternative sidesteps the fragile shell mechanics entirely: amplifying the (already present, but weak) activator-dependent *division* into a sharp $\sim$4:1 ON/OFF contrast on the fast, stable spherical mechanics produces *localised* extrusion that tracks the morphogen ($\mathrm{corr}(a, r) = 0.31$ vs. ≈ 0 for the shell), while the monolayer stays clean – the high-activator domains bulge out into lobes.

1 The model in one paragraph

One `cell` set carries three independently integrated state blocks: position `pos` (the vertex coordinate), morphogens `chem` $= [a, h]$ (a Gray–Scott or Brusselator RD on the cell–cell adjacency graph, so mechanical deformation reshapes the diffusion), and a target size `v0` that a Hill-saturating growth law inflates. Division wakes a dormant buffer slot when `v0` reaches its threshold. Mechanics is an overdamped shape energy $U = \sum_j K_V(V_j - V_j^0)^2 + K_S S_j$ over ghost-bounded Voronoi cells; the force is $-\nabla U$ by autodiff through the tetrahedral circumcentres. Each of these is a `plexus2` operator (a typed morphism on the set) composed by a schedule; different operators run at different cadences, which is how the multiple timescales are expressed.

2 What transferred cleanly: Fig. 4 (patterns on a growing vesicle)

In Fig. 4 growth is *activator-independent* (homogeneous), so the RD only colours the cells. The naive Gray–Scott vesicle looked badly corrupted — a lumpy, non-epithelial surface — and it was tempting to blame the reaction–diffusion. It was not: the mechanics was simply being integrated at Gray–Scott's coarse $\Delta t = 0.35$, far outside the vertex model's stability window. The fix is to *decouple the timescales*: keep the mechanics at the fine step that makes it clean ($\Delta t = 0.02$, the same parameters that produce a nice Brusselator vesicle) and speed the RD up by a rate multiplier ρ (with the diffusion coefficient scaled to match), so the identical pattern develops in the same number of frames. Roughly $\rho \approx \Delta t_{\mathrm{coarse}}/\Delta t_{\mathrm{fine}}$; the only ceiling is the explicit-diffusion CFL limit ($\Delta t\, d_h\, \chi\, \bar{k} \lesssim 0.4$), which the delicate low-feed labyrinth regime hits first and which must be respected per regime. The result (Fig. 1) is a uniform epithelium carrying the branching coral pattern, with shell roughness 0.04 against 0.24 for the coarse-step version. The take-away is general: *when a signal does not feed back on shape, do not let its integration timescale dictate the mechanics'.*

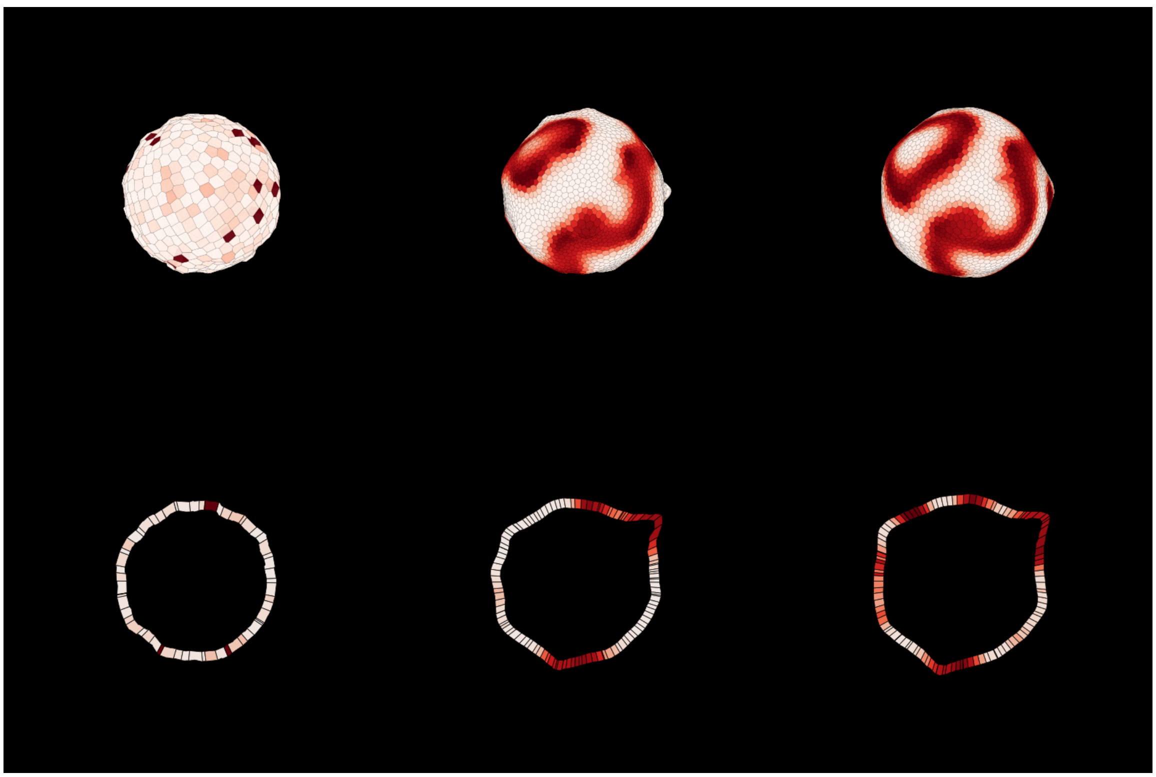

Figure 1: **Gray–Scott coral on the vesicle mechanics** (`fig4_coral_v`), at 0/50/100% of the run. Top: closed external view; bottom: cross-section through the monolayer and lumen. Cells are uniform (near-hexagonal) and the shell is smooth; the reaction–diffusion is a pure colouring of a clean epithelium. $N\!: 600 \to 3000$, roughness 0.04.

3 A cautionary tale: the silent no-op

The uniform cells above depend on a tangential (Lloyd) relaxation that moves each cell toward its spherical-Voronoi centroid. That operator had *never once* executed: the module imported `Delaunay, ConvexHull, cKDTree` but not `SphericalVoronoi`, so every call raised `NameError`, which fell into a blanket `except Exception: return zeros`. The knob was inert — `k_lloyd` 0.3/1.0/2.0 gave byte-identical output. The uniformity previously visible came entirely from a *render-side* Lloyd pass (a cosmetic applied only when drawing). The moment we honoured "uniformity should come from physics, not the render" and drew the true cells, the dead operator was exposed. Two lessons: a broad `except` around a geometry call can hide a programming error for the lifetime of a component; and a cosmetic post-process can mask a missing mechanism indefinitely. The fix was one import plus narrowing the handler to the genuine Qhull degeneracies; the operator was also made timestep-independent (move a fixed *fraction* toward the centroid per tick, capped at 0.7× the cell spacing).

4 What did not transfer: Fig. 5 (tubulation)

Fig. 5 makes the activator a *mitogen*: high-activator cells grow, and on a closed sheet the locally excess area must buckle out of plane into tubes — no explicit extrusion force. We implemented the coupling (a Hill switch from activator concentration to target-volume growth) on the apical/basal "shell" mechanics, and swept ~ 15 configurations of domain size, growth rate, bending, surface tension, and lumen pressure. *None* produced clean tubes. The behaviour is bimodal: gentle growth or stiff bending gives a perfectly smooth sphere (no buckling), while enough growth gives *global* chaotic buckling that tears the monolayer into a blob. Decisively, the correlation between activator concentration and radial protrusion is ≈ 0 in every case: the bulges are *not* where the activator is — it is generic Euler buckling of an over-grown, lumen-pressurised shell, not localised extrusion.

Three mechanics ingredients are missing, all present in the original: (i) *force-balance iteration* — Okuda solves the overdamped balance to residual $< 10^{-5}$ each Δt_v, whereas a single explicit step lets individual cells spike rather than forming coherent folds; (ii) *topological reconnection* (RNR / T1), without which cells jam instead of flowing into a tube; and (iii) a tractable per-cell tessellation — our `ConvexHull`-per-cell shell becomes pathologically slow, even hanging, exactly in the heavily-deformed regime tubes require. Clean tubulation is thus gated on a vectorised, force-balance-iterated shell mechanics plus a reconnection operator, not on more parameter tuning.

5 Field notes: what we learned about the model

Grouped by subsystem. The numbers are the values that actually worked — or failed — in this prototype; they are the hard-won part and the main reason for this document.

Timescales — the master knob.

- Gray–Scott needs a large step ($\Delta t \approx 0.35$) to develop in $\sim 10^4$ frames; the overdamped vertex model is stable only for $\Delta t \lesssim 0.05$. Irreconcilable at a single step.
- Resolution: run mechanics at $\Delta t = 0.02$ and multiply the RD rate by $\rho \approx \Delta t_{\text{coarse}}/\Delta t_{\text{fine}} \approx 17.5$, scaling the diffusion prefactor χ by the *same* ρ so the reaction:diffusion ratio — and thus the pattern — is preserved. A pure time-rescaling: identical pattern, stable mechanics.
- Explicit graph-diffusion is stable only while $\Delta t\, d_h\, \chi\, \langle k \rangle \lesssim 0.4$. With $d_h = 0.16$, $\langle k \rangle \approx 14$, $\Delta t = 0.02$ this caps $\chi \lesssim 9$. Robust high-feed coral ($F = 0.058$) tolerates $\chi = 9.8, \rho = 35$; the delicate low-feed labyrinth ($F = 0.029$) *diverges to NaN* there and needs $\chi = 5.6, \rho = 20$ over more frames. Tune the rescaling *per regime.*

Tangential (Lloyd) relaxation.

- Make it timestep-independent: move a fixed fraction `k_lloyd` of the way to the spherical-Voronoi centroid per tick, then divide by Δt to undo the engine's $x \mathrel{+}= \Delta t\, v$. `k_lloyd` is then a clean fraction (1.0 = one full Lloyd step/tick; 1.5 slightly overshoots and converges faster). Cap the per-tick move at 0.7× mean spacing so a defect cell cannot jump neighbours and tangle the mesh.
- It uses `SphericalVoronoi` of the cell *direction* vectors, which degenerates once the surface is non-spherical — so it is valid on the vesicle, *not* on Fig-5 tubes.

Morphogen → growth (the coupling).

- $\lambda = \lambda_{\mathrm{ref}}\big(\rho + g\,\mathrm{Hill}(a)\big)$, $\mathrm{Hill}(a) = a^{\alpha}/(a^{\alpha} + a_{\mathrm{sw}}^{\alpha})$. ρ is the OFF multiple, $\rho + g$ the ON one.
- a_{sw} *must* sit inside the activator's actual range or the switch never fires. The Gray–Scott activator peaks near 0.4, so $a_{\mathrm{sw}} = 1.0$ gives Hill$\approx$0.16 (a feeble 16% contrast); $a_{\mathrm{sw}} \approx 0.2$ makes it bite. $\alpha = 4$ gives a near-step switch.
- Even "homogeneous" Fig-4 growth ($\rho = 1$) carries the Hill term, so high-activator domains already bulge weakly. Amplify with g ($\rho = 0.5, g = 1.5 \Rightarrow 4{:}1$) for real localised proliferation.

Division.

- `divide_2x` fires when v^0 reaches `ratio`×base ($v_{\mathrm{th}} = \frac{4}{3} v_{\mathrm{ref}}$); the daughter wakes a dormant buffer slot, both reset to base, and the daughter inherits the mother's `chem`. Faster growth ⇒ earlier division ⇒ local density ⇒ local area excess — the buckling driver.
- `reset_noise` spreads daughter target sizes so the cycle stays asynchronous (no division waves). Fixed-buffer occupancy: division stops when the buffer is full.

Mechanics — two flavours; know which.

- *Spherical-ghost* `voronoi_tension_3d`: fast (vectorised, ~32× over the per-cell loop), stable, keeps a clean monolayer — but the global ghost sphere *fights* local deformation, so at most mild bumps. Right for Fig 4 and the differential-growth route.
- *Per-cell apical/basal shell* `voronoi_tension_shell`: allows local buckling but needs a scipy `ConvexHull` per cell; under heavy deformation this is pathologically slow and can *hang* (observed 77-min hang, 84-min no-finish). Not viable for tubes un-vectorised.
- Stiff volume vs. surface, $K_V/K_S \approx 10$ (incompressible cells). Bending k_{bend} HIGH (0.6–0.8) forbids out-of-plane curvature → no buckle; $k_{\mathrm{bend}} \approx 0.45$ permits it.

Fig-5 buckling phenomenology (all negative, all informative).

- Bimodal: gentle growth / stiff bending → smooth sphere; enough growth → global chaotic buckle that tears the monolayer. No clean-tube window in between.
- `lumen_pressure` holds the closed vesicle rigid, so area is stored until it releases as one global Euler buckle at a random site — morphology decoupled from the pattern. Removing the lumen did *not* localise it ($\mathrm{corr}(a, r)$ stayed ≈ 0).

Diagnostics that earned their keep.

- *roughness* = std/mean of cell radii (0.04 clean, 0.24 corrupted); *% hexagons* (closed sphere tops out $\sim 70\%$; a bounded 2D disc lower, its boundary ring never hexagonal); *PROTR* = 99th-percentile/median radius (≈ 1 sphere, > 1.4 protrusions); $\mathrm{corr}(a, r)$ = the decisive extrusion-is-localised test.

Performance / engineering.

- Multi-rate composition via an engine `every:K` gate plus an intra-operator topology cache (`retess_every`): mechanics reuses the tessellation for K ticks, pattern preserved, $\sim 1.7\times$ at $K = 10$.

- `torch.compile` gives $\sim 50\times$ on the fixed-N energy but recompiles when N grows (division) — keep it off for growing runs. CPU/scipy is the real bottleneck; on the cluster, one dedicated node per generation ($\sim 3\times$ over contended local runs).

6 Discussion: a cleaner route to extrusion

The negative result above is informative. Localised growth on a *closed, pressurised* vesicle tends to store area until it releases as a single global buckle at a random site — morphology decoupled from the pattern. A better substrate for *local* extrusion is the fast, stable *spherical* mechanics used for Fig. 4 (it never breaks the monolayer), driven by *localised proliferation* rather than global inflation.

Strikingly, that mechanism is already latent in the Fig. 4 runs. Even "homogeneous" growth carries a Hill term, $\dot{v}^0 = \lambda_{\mathrm{ref}}\big(\rho + \mathrm{Hill}(a)\big)$, so high-activator cells already divide somewhat faster and their domains bulge slightly outward — the small extrusions one can see in the coral cross-section. They are weak only because the switch was mis-set ($a_{\mathrm{sw}} = 1.0$ while the Gray–Scott activator peaks near 0.4, so Hill ≈ 0.16). Introducing a gain, $\dot{v}^0 = \lambda_{\mathrm{ref}}\big(\rho + g\,\mathrm{Hill}(a)\big)$ with $\rho = 0.5$, $g = 1.5$ and a_{sw} moved into the activator's actual range, turns this into a sharp 2×-ON / 0.5×-OFF (4:1) *division* contrast that concentrates proliferation — and hence out-of-plane buckling — at the morphogen domains, on mechanics that stay stable and cheap.

This works (Fig. 2). The high-activator domains bulge outward into coherent lobes, the monolayer stays intact (roughness 0.058, cells 54% hexagonal), and crucially the deformation now *tracks the signal*: $\mathrm{corr}(\mathrm{activator}, \mathrm{radius}) = 0.31$, against ≈ 0 for every shell-mechanics attempt. It is not yet the fully elongated tubes of the original — for that the force-balance and reconnection ingredients of §5 are still needed — but it is genuine, localised, morphogen-driven extrusion on a substrate that never breaks. This reframes tubulation as *patterned proliferation* on a robust substrate rather than differential growth on a fragile deformable shell.

Can the lobes be elongated? (a cluster sweep). We ran a parameter sweep on the L4 cluster to push the lobes toward tubes. Two findings. *(i) "Run longer" is the wrong lever, and it is dangerous.* Letting the tissue over-grow (bigger buffer, more frames) both *relaxes* the protrusions back to a sphere once growth saturates ($\mathrm{protr}_{\mathrm{max}}$ falls $1.27 \to 1.08$, corr $\to 0$) and, worse, packs cells so densely at the activator domains that the explicit-diffusion *CFL is violated* — the RD blows up to NaN (a silent failure we now catch with an `act_nan` flag). The extrusion lives in a finite $N \sim 3000$ window. *(ii) Sharper division contrast elongates — but toward spikes.* At the productive scale, raising the ON/OFF contrast from 4:1 to ~ 6.7:1 ($\rho = 0.3, g = 1.7$) is the only

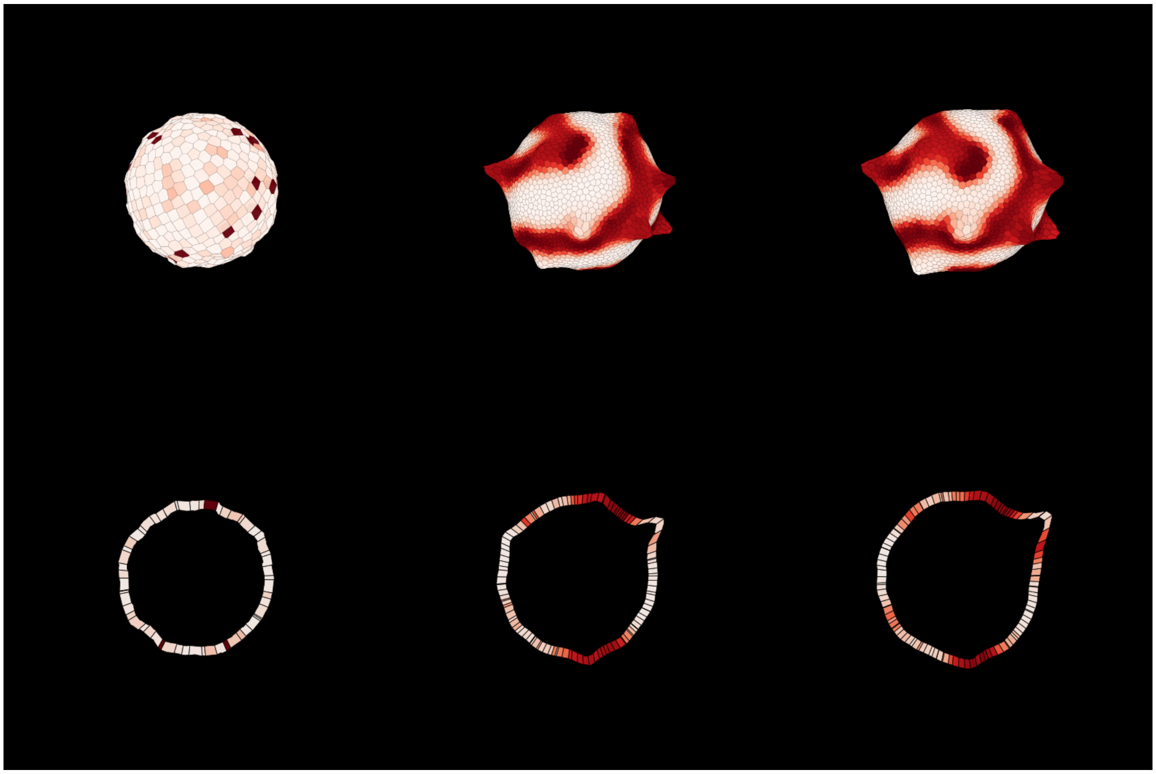

Figure 2: **Locally-favoured extrusion** (`fig4_coral_ext`): the same coral RD on the fast spherical mechanics, but with a 4:1 activator-driven division contrast ($\rho = 0.5, g = 1.5, a_{\mathrm{sw}} = 0.2$). The high-activator (dark) domains extrude outward into lobes while low-activator regions stay flat; the cross-section (bottom) shows the monolayer lobing out where the signal is high, and remaining a clean single sheet. $\mathrm{corr}(a, r) = 0.31$.

lever that clearly extends the tips ($\mathrm{protr}_{\max} = 1.46$, the tallest we reached), and the tips stay red (activator-led). But Fig. 4 shows what "taller" means here: the broad lobes sharpen into pointed *fingers* — individual cells and small clusters shooting out — rather than coherent tubes. Lowering surface tension ($K_S = 0.4$–0.25) helps mildly ($\mathrm{protr}_{\max} \approx 1.2$–$1.27$). Nor does bending rescue it: adding bending smooths the spikes but *flattens the whole extrusion* — both $\mathrm{protr}_{\max}$ and corr fall ($1.44 \to 1.36$, $0.23 \to 0.07$). Across the whole sweep the elongation *plateaus at* $\mathrm{protr}_{\max} \approx 1.44$: you cannot get tall, clean, and localised at once on this mechanics — push contrast and it spikes, add bending and it flattens. This is the same wall as §5 from the other side: the fast spherical SPV mechanics can *localise* growth beautifully (best balanced config `ext3_contrast`: $\mathrm{protr}_{\max} = 1.44$, $\mathrm{corr} = 0.23$, roughness 0.039) but cannot hold a coherent extended neck. Clean elongation is gated on the coherent vertex mechanics (shared vertices + force balance + T1), i.e. a true vertex model such as Tyssue — the natural next substrate.

Dividing directly on activation. The cleanest coupling of all is to divide cells *directly* where the activator is on, rather than routing through growth-then-size (a one-line `divide_2x` option). The subtlety is that the per-tick division probability must be *graded* in the activator, $p = \texttt{act\_rate} \cdot a^n/(a^n + a_{\mathrm{thr}}^n)$: a hard threshold lets pale cells that merely exceed it divide at the *full* rate, so low-activator cells proliferate too (a flaw we caught by eye and fixed). With the graded, sharply-gated rate (`ext5_sharp`, $a_{\mathrm{thr}} = 0.35$, $n = 6$), proliferation is confined to the deep-activator cells — Fig. 3

shows the signature directly in the cell *sizes*: the activator (dark) bands fill with small, densely packed daughters and bulge outward, while the low-activator regions keep their large original cells and stay flat. This is the most *pattern-faithful* extrusion of the whole study, $\mathrm{corr}(a, r) = 0.42$ — well above the best growth-driven config (0.23) and the broad-lobe `coral_ext` (0.31) — on a clean monolayer. The stable window is narrow (combining direct division *with* growth, or too high a rate, over-proliferates and blows the RD CFL to NaN). So direct, activation-graded proliferation is the tightest pattern→morphology coupling available on this mechanics; the residual neck-coherence wall still caps the *height* ($\mathrm{protr}_{\max} \approx 1.4$), which is what the vertex model would relieve.

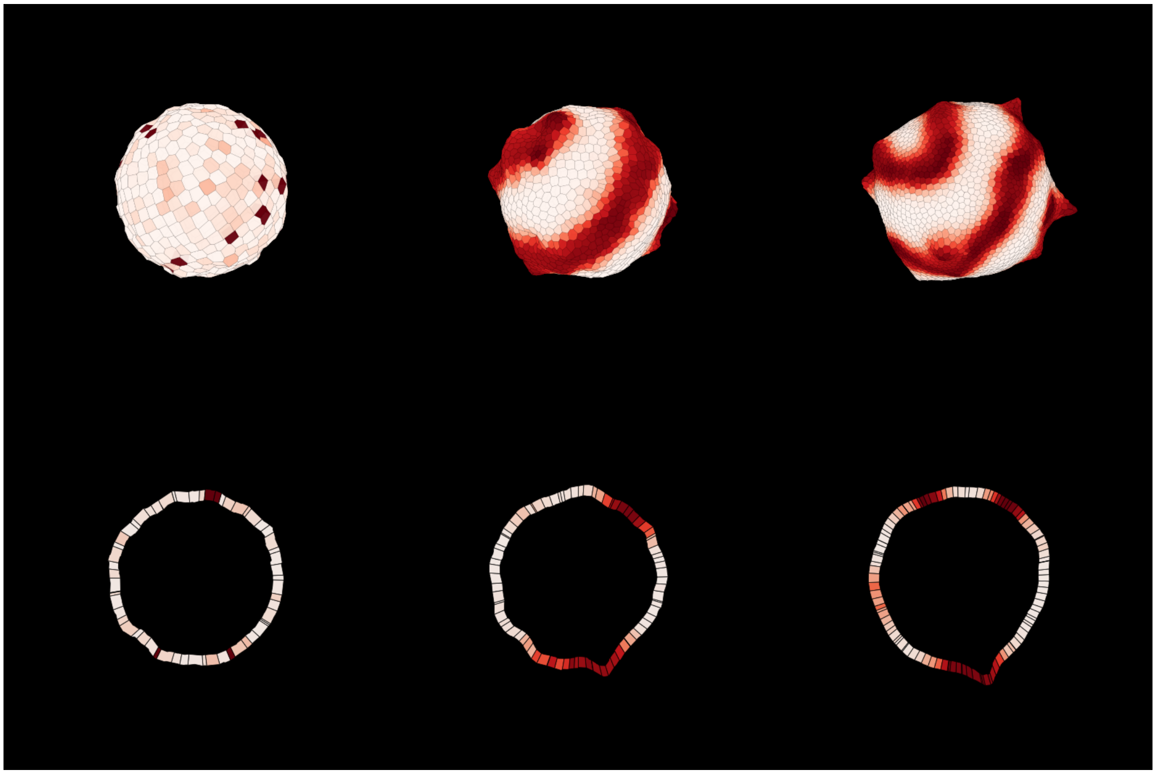

Figure 3: **Divide-on-activation, activator-graded** (`ext5_sharp`): division probability $\propto$ Hill(activator), no growth. Proliferation is confined to the activator (dark) bands — read it off the cell sizes: dark bands are small, dense divided cells that bulge out; pale regions keep their large, undivided cells and stay flat. The most pattern-faithful extrusion of the study, $\mathrm{corr}(a, r) = 0.42$, on a clean single-sheet monolayer (cross-section, bottom).

Two methodological points recur. First, *signalling is geometry-coupled but not mechanics-coupled unless you say so*: keeping that boundary explicit turned an intractable "noisy patterning" problem into a one-line timescale fix. Second, in a typed-operator language the right unit of change is often a new *implementation* or *parameter* of an existing contract (the 4:1 growth is still the `growth` operator), not a new operator — "prototype freely, promote mechanisms conservatively."

7 The gap to Figure 5, and the Tyssue bridge

This prototype set out to reach Okuda's tubulation (Fig. 5) and branching (Fig. 6) and did not — but the failure is now precisely located, and a parallel prototype (`prototype/Tyssue`) is already most of the way across the gap.

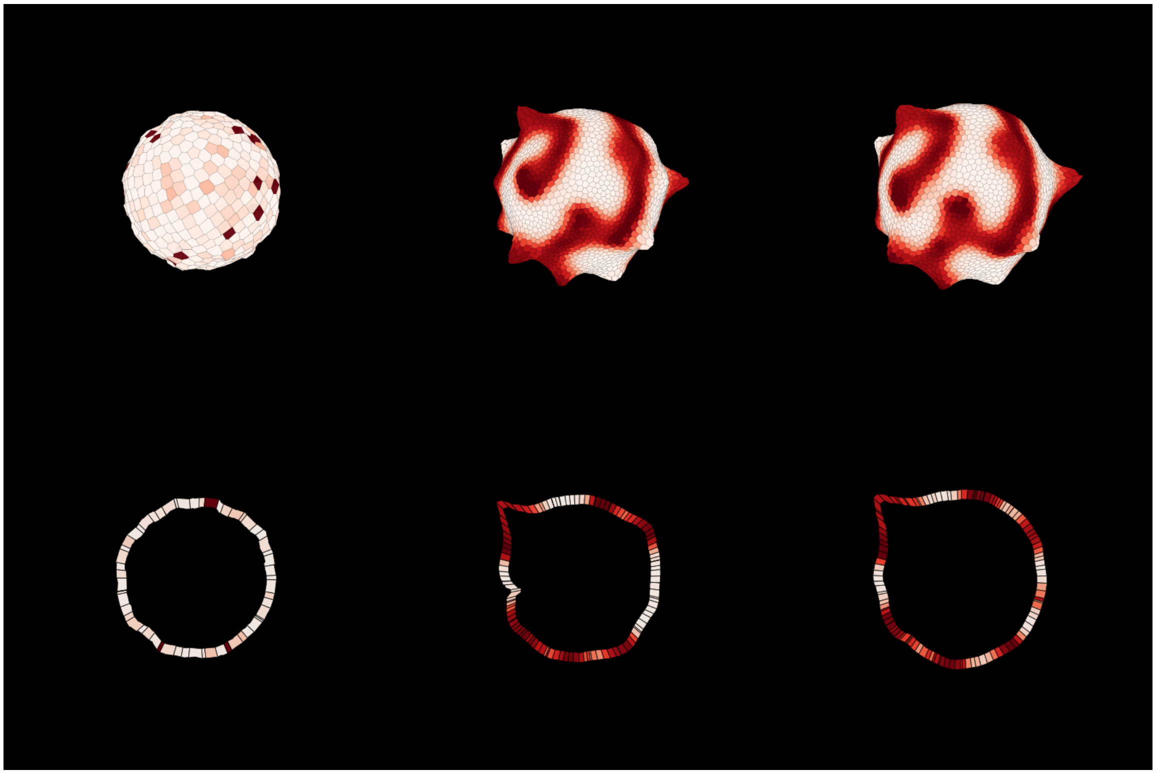

Figure 4: **Elongation by sharper contrast** (`ext2_contrast`, 6.7:1): the tallest protrusions we reached ($\mathrm{protr}_{\mathrm{max}} = 1.46$), tips still activator-led (red). But the extrusions sharpen into pointed *fingers*/spikes rather than coherent tubes — the signature of driving growth harder on a mechanics that lacks shared vertices and force balance.

The gap, in the language of the model ladder. Confluent-tissue mechanics forms a ladder of increasingly permissive representations, distinguished by *what the degrees of freedom are*. At the bottom sits the *Self-Propelled-Voronoi* (SPV) model we used: the only DOF are the cell *centres*, and each cell's shape is not stored but *read off*, every step, as the Voronoi region of the centres (Bi 2016; Sussman 2017). One rung up is the *vertex model*, whose DOF are the shared tri-cellular *junctions*: cells are generic polygons with explicit edges and line tensions, free to take shapes no Voronoi partition admits. At the top is the full 3D vertex / deformable-cell model (Okuda 2018; SimuCell3D), where each cell is a polyhedron with an apical (lumen-facing) and basal surface, an enclosed volume, and a dynamically reconnecting network.

A tube is exactly what the bottom rung cannot hold. It is (i) a high-aspect-ratio junctional network far from any Voronoi tessellation of the centres — which the SPV forces the geometry back onto at every step; (ii) an intrinsically 3D object with two principal curvatures and an apical/basal axis — which a planar point-tessellation has no coordinates for; and (iii) defined by an enclosed *lumen* — of which the SPV has no notion. This is why our extrusion *localised* superbly (graded divide-on-activation, $\mathrm{corr} = 0.42$) but its *height* plateaued at $\mathrm{protr}_{\mathrm{max}} \approx 1.4$: with the shape slaved to the centres and no way to store a coherent neck, driving growth harder gave single-cell spikes, not tubes. *The wall is representational, not parametric.*

What Okuda's tube actually needs — four coupled ingredients, none native to the SPV. *(1) Quasi-static force balance:* the overdamped vertex equations solved to residual $< 10^{-5}$ *between* growth steps (Eq. 1; $\tau_{\text{cycle}} = 50 \gg \Delta t_v = 0.005$). *(2) Incompressible cells + surface tension:* $U = \sum_j \frac{1}{2} k_v (v_j - v_{\text{eq},j})^2 + \kappa_s s_j$, $k_v = 10 \gg \kappa_s = 0.2$ (Eq. 3) — growth of a cell's target volume must be absorbed by shape change, not compression. *(3) Reversible network reconnection (the 3D T1):* any edge below $\Delta l_{\text{th}} = 0.05$ reconnects, attempted every $\Delta t_r = 1.0$ — the topology change that lets a bump *flow* into a tube. *(4) Activator→growth:* a discrete Gierer–Meinhardt mitogen (Hill $\alpha = 10$ step at $\rho_{\text{sw}} = 0.5$, Eqs. 7–8), the spot self-maintaining at the growing tip. Tube diameter tracks the steady domain size $\propto \chi^{1/4} \phi^{1/2}$ (thick: $\chi = 0.1, \gamma = 1$; thin: $\chi = 0.01, \gamma = 100$).

Tubes, branches, undulations are one knob apart. The *same* growing vesicle gives undulation, tubulation, or branching depending only on the ratio of patterning rate γ to deformation rate $1/\tau_{\text{cycle}}$ — the *hysteresis* of the Turing pattern on the growing domain. $\gamma = 100$ (pattern outruns growth) → domains wander → *undulation* (Fig. 7); $\gamma = 1$ (domains held in place) → persistent tip zones → *tubes* (Fig. 5); $\gamma = 0.01$ with small domains ($\chi = 0.01$) → division-induced morphogen dilution repatterns a spot into separated sub-spots → *branching* (Fig. 6). The last is a genuinely *discrete-cell* effect: because the RD is homogeneous *per cell*, growth/division inject a cell-scale discontinuity that, under high hysteresis, seeds the branch — precisely where a discrete vertex model does work a continuum cannot.

Rigidity is the precondition. Whether a tissue *can* tubulate is set by the shape-index transition (Bi 2015): at $p_0 = P_0/\sqrt{A_0} \approx 3.81$ the T1 barrier vanishes and the sheet fluidises. Below it the sheet jams (growth builds stress but the wall cannot flow); above it, it flows. Okuda does not tune p_0 — he sets the reconnection *cadence* ($\Delta t_r \ll \tau_{\text{cycle}}$) so the tissue is quasi-statically fluid. Two ratios thus select morphology: reconnection-vs-growth (must be fluid to tube at all) and patterning-vs-growth (undulate/tube/branch). Our SPV had *neither* knob — no explicit T1, no shape-index handle — the same gap seen from the rigidity side.

The Tyssue bridge (parallel prototype). We built `prototype/Tyssue`, a true vertex model (AVM) as `plexus2` operators, expressly to supply the two ingredients the Voronoi route lacks — force-balance iteration and explicit T1 — under the *same* shape-energy contract. It is already well across the gap. In 2D it reproduces the $p_0^* \approx 3.81$ rigidity transition from genuine force balance (residual energy collapses ~ 4 orders across p_0^*, no timestep tuning — the SPV's "master-knob" problem gone), with validated T1, oriented division, apoptosis, growth and RD operators. In 3D it already carries a closed epithelial *vesicle* half-edge mesh (`seed_mesh_3d`), a 3D shape energy with a *per-cell lumen* volume (`shape_energy_3d` — the incompressibility + surface tension of Eq. 3), 3D cell *division* (`divide_3d`), quasi-static growth, and a `vesicle_fig4` preset reproducing our Fig. 4 growth scale on the true mesh.

The continuing steps to Fig. 5, then Fig. 6. Against Okuda's ingredient list, Tyssue needs, in order:

1. **A 3D reconnection — the one genuinely missing mechanics op.** A 3D T1 / IH–HI reconnection on the vesicle mesh (only the 2D T1 exists today), triggered below an edge-length threshold on a slow cadence ($\Delta t_r \ll \tau_{\text{cycle}}$). This is what lets a bump *extend* into a tube rather than jam; Barton's continuous shrink-to-point edge flip is the robust recipe if a discrete threshold proves brittle under large deformation.

2. **Couple the RD to the 3D mesh.** Tyssue's RD runs in 2D today and the vesicle grows *uniformly*; run the discrete Gierer–Meinhardt (or Gray–Scott) on the vesicle's cell adjacency,

flux $\propto$ (shared face area)/(cell volume) so pattern and geometry co-evolve (Eq. 2), reduced to the two knobs χ (diameter) and γ (hysteresis).

3. **Port the winning growth coupling onto the mesh.** Carry over exactly what won here — activator-*graded* proliferation ($\dot{v}_{\rm eq} \propto \mathrm{Hill}(a)$, or divide-on-activation at the graded rate, corr $= 0.42$) — with oriented (Hertwig) division. On the true mesh, force balance and T1 should convert the SPV's spiky fingers into coherent constant-diameter tubes. *With these three, the prototype should tubulate.*

4. **Sweep** (χ, γ)**.** Reproduce the morphology diagram (Fig. 8) on a $\sim$2000-cell vesicle: $\gamma = 100 \rightarrow$ undulation, $\gamma = 1 \rightarrow$ tube, $\gamma = 0.01, \chi = 0.01 \rightarrow$ branching; verify the $\chi^{1/4}$ diameter scaling.

5. **Preserve per-cell discreteness for branching.** Keep the RD field homogeneous *per cell* and run high hysteresis, so division-induced dilution injects the cell-scale perturbation that splits a domain into a branch (Fig. 6).

The lumen is already in `shape_energy_3d`; membrane bending can be added later to set thin-tube undulation vs. thick-tube straightness. Caveat: the whole ladder assumes quasi-static mechanics and slow growth — if real tip-driven branching (traction, motility) is later needed, the active machinery of the AVM/SPV (Barton 2017; Bi 2016) is the natural next layer on the 3D geometry.

In sum. The Turing$\times$vertex *coupling* was never the hard part — Fig. 4 shows a clean epithelium carrying any Turing pattern, and the extrusion experiments show morphogen-graded proliferation localises deformation faithfully (corr $= 0.42$). The hard part is the *mechanics substrate*: tubes and branches need shared junction vertices, an enclosed lumen, quasi-static force balance, and reversible reconnection. The SPV cannot provide them; the parallel Tyssue vertex model already provides most, and is *three well-defined operators* — a 3D T1, an RD-on-mesh coupling, and the graded-growth port — from Fig. 5, and *one regime knob* (γ) beyond that from Fig. 6.